

KAZI FAHIM AHMAD NASIF

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Portfolio: kazi-nasif.github.io LinkedIn: [kazi-fahim-ahmad-nasif](#)

Google Scholar: [Kazi Fahim Ahmad Nasif](#)

Research Interests

Bioinformatics, World Models, LLMs, Computer Vision, Edge Computing.

Education

Ph.D. in Computer Science

08/2023 – Present

Kennesaw State University, Marietta, Georgia, USA

Passed Ph.D. candidacy – Spring 2026

MS in Applied Statistics & Data Science

09/2019 – 10/2021

Jahangirnagar University, Dhaka, Bangladesh

BEng. in Computer Science & Engineering

04/2014 – 09/2019

Rajshahi University of Engineering and Technology (RUET)

Research Experience

Graduate Research Assistant

08/2023 – Present

Kennesaw State University

- **Bioinformatics:** Developed tools for accelerating the Molecular Dynamics and Protein-Protein Docking with faster inference and low computational resources.
- **World Model:** Developed energy efficient frameworks for meteorological events using world model. These solutions are deployable in resource-constrained edge devices.
- **Physical AI:** Developed multimodal computer vision system with custom dataset, capable of real-time inference with local models which is secured.
- **Explainable AI:** Built a robust and secure AI decision-support system for critical applications under uncertainty.

- **NGO funded project (*Food First*)** ([Link](#))

08/2024 – Present

Developed scalable data management and storage solutions, integrated customized AI tools for website management, traffic analysis, and recommendation-driven user preference modeling, and guided graduate students in applying AI to improve website performance.

- **Industry funded project (*InComm Payments*)** ([Link](#))

08/2024 – 07/2025

Built compressed, edge-deployable multimodal vision systems for shopper behavior analysis, covering emotion detection, gaze estimation, product detection, and hand-interaction tracking to support planogram design and shelf optimization. This solution can be applied in patient monitoring also.

Teaching Experience

Guest Lecturer

Fall 2026

IT 3503 – Foundations of Health Information Technology, Kennesaw State University

- Delivered a guest lecture on “How to Use AI Ethically and Safely for Academic Research in Clinical Decision Support Systems” to undergraduate IT students.

Professional Experience

Actuarial Analyst

11/2021 – 07/2023

Guardian Life Insurance Limited

- Conducted business forecasting by analyzing historical data and market conditions for long-term planning.
- Performed financial risk management for budget optimization.
- Executed business valuation by analyzing financial statements, future earnings, and market dynamics.
- Developed rate charts using life tables and actuarial models for premium calculations.

Trainee Programmer

03/2019 – 06/2019

Sysnova Information Systems

- Customized LibreOffice Calc functionalities by developing macros to automate tasks.
- Enhanced data visualization and reporting features for organizational workflows.
- Prepared documentation and training materials to support adoption of customized tools.

Publications

- **Kazi Fahim Ahmad Nasif***, Madeline Frank, Md Majedul Islam, Pritam Dutta, Md Romyull Islam, Priyabrota Sen, Ramazan Aygun, Bobin Deng. *MM-JEPA-TC: Low Cost Multimodal Joint-Embedding Predictive Architecture for Tropical Cyclone Forecasting*, 2026 IEEE International Conference on Systems, Man, and Cybernetics (IEEE SMC 2026). Accepted.
- Lingtao Chen, Shuteng Niu, Zhiyu Dai, Xiao Luo, Chen Zhao, **Kazi Fahim Ahmad Nasif**, Yixin Xie. *AffinityBERT: Sequence-Based Deep Learning for Structure-Free Protein Binding Prediction*, 2026 IEEE International Conference on Systems, Man, and Cybernetics (IEEE SMC 2026). Accepted.
- Bobin Deng, **Kazi Fahim Ahmad Nasif**. *Lowering Inference Latency of Hyperdimensional Computing for Camera Frame Inputs via Pixel Differential Compression*, 2026 8th International Conference on Artificial Intelligence Technologies and Applications (ICAITA 2026). Accepted.
- Md Majedul Islam, **Kazi Fahim Ahmad Nasif**, Nobel Dhar, Mahmut Karakaya, Selena He. *Extreme Parameter Efficiency in Quantum Machine Learning for Cervical Cancer Classification*, 48th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (IEEE EMBC 2026). Accepted.
- L. Chen*, Y. Deng, Z. Dai, S. Niu, C. Tao, **K. F. A. Nasif***, B. Deng, Y. Xie. *Mechanistic Validation and Structural Analysis of PAH-Associated SOX17 for Real-World Relevance*, 14th IEEE International Conference on Healthcare Informatics (IEEE ICHI 2026). [Link](#)
- **Kazi Fahim Ahmad Nasif**, Bobin Deng, Yixin Xie, Syed Md Shamsul Alam, Lingtao Chen, Nobel Dhar, Kun Suo, Dan Chia-Tien Lo. *AccelMD: A Self-adaptive AI-enabled Framework for Accelerating Molecular Dynamics Simulations*, 37th IEEE International Conference on Tools with Artificial Intelligence (ICTAI 2025). [Link](#)
- L. Chen, **K. F. A. Nasif**, M. Yang, S. Niu, B. Deng, Y. Xie. *Automatic Molecular Dynamics Simulation with Binding Affinity Prediction Using Deep Learning*, 37th IEEE International Conference on Tools with Artificial Intelligence (ICTAI 2025). [Link](#)
- Z. Elghazzali, Y. Xie, C. Zhao, B. Deng, L. Chen, **K. F. A. Nasif**. *Multimodal Deep Learning for Alzheimer's Disease Classification: A Practical Fusion Framework for Clinical Deployment*, 37th IEEE International Conference on Tools with Artificial Intelligence (ICTAI 2025). [Link](#)
- Md Nurul Absur, Sourya Saha, Sifat Nawrin Nova, **Kazi Fahim Ahmad Nasif**, Md Rahat Ul Nasib. *Optimizing CDN Architectures: Multi-Metric Algorithmic Breakthroughs for Edge and Distributed Performance*, International Conference on Computing, Networking and Communications (ICNC 2025). [Link](#)
- Nobel Dhar, Bobin Deng, Md Romyull Islam, **Kazi Fahim Ahmad Nasif**, Liang Zhao, Kun Suo. *Activation Sparsity Opportunities for Compressing General Large Language Models*, 43rd IEEE International Performance Computing and Communications Conference (IPCCC 2024). [Link](#)
- M. N. Absur, **K. F. A. Nasif**, S. Saha, S. N. Nova. *Revolutionizing Image Recognition: Next-Generation CNN Architectures for Handwritten Digits and Objects*, IEEE Symposium on Wireless Technology & Applications (ISWTA 2024). [Link](#)
- L. Chen, Q. Li, **Kazi Fahim Ahmad Nasif**, Y. Xie, B. Deng, S. Niu, S. Pouriyeh, Z. Dai, J. Chen, C. Y. Xie. *AI-Driven Deep Learning Techniques in Protein Structure Prediction*, *International Journal of Molecular Sciences*, 25(15):8426, 2024. [Link](#)
- Lingtao Chen, **Kazi Fahim Ahmad Nasif**, Bobin Deng, Shuteng Niu, Chloe Yixin Xie. *Predicting Protein-Protein Binding Affinity with Deep Learning: A Comparative Analysis of CNN and Transformer Models*, 36th IEEE International Conference on Tools with Artificial Intelligence (ICTAI 2024). [Link](#)
- **K. F. Ahmed Nasif**, M. Nurul Absur, M. Al Mamun. *Order Dependency in Sequential Correlation*, 3rd International Conference on Electrical, Computer & Telecommunication Engineering (ICECTE 2019). [Link](#)

In Progress

- **Kazi Fahim Ahmad Nasif**, Lingtao Chen, Yixin Xie, Bobin Deng. *DockGraph: Fast Protein-Protein Docking Refinement with a Dual-Encoder Graph Neural Network.*, Submitted to CIKM 2026.
- **Kazi Fahim Ahmad Nasif**, Lingtao Chen, Priyabrota Sen, Yifan Liu, Michael Sierks, Bobin Deng, Yixin Xie. *BEYOND POINT-WORLD REASONING: CERTIFIED DECISIONS UNDER UNCERTAIN PERCEPTION*, Submitted to ICASSP 2027.
- **Kazi Fahim Ahmad Nasif**, Lingtao Chen, Priyabrota Sen, Yifan Liu, Kewei Chen, Bobin Deng, Yixin Xie. *Marginal Coverage Is Not Enough: Evaluating Conformal Prediction in Ordinal Medical Imaging*,

Submitted to ICASSP 2027.

- Lingtao Chen, **Kazi Fahim Ahmad Nasif**, Yifan Liu, Bobin Deng, Yixin Xie. *AffinitySym: Sequence-based Interpretable Protein-Protein Binding Affinity Prediction*, Submitted to ICLR 2027.
- Priyabrota Sen, Bobin Deng, **Kazi Fahim Ahmad Nasif**, Patrick Bobbie, Liang Zhao, Sahidul Islam, Kun Suo, Yixin Xie. *DOR-JEPA: Difficulty-Ranked Slot Masking for Causal World Model Learning*, Submitted to ICASSP 2027.
- Priyabrota Sen, Bobin Deng, **Kazi Fahim Ahmad Nasif**, Jiajing Huang, Kun Suo. *A Vision-Driven Decision-Enabled Closed-Loop Digital Twin for Smart Production Lines*, Submitted to ICRA 2027.

Patent

- Bobin Deng, **Kazi Fahim Ahmad Nasif**, Nobel Dhar, Rahim Snow, George Harbison, Michael Parlotto. *Systems and Methods for Using Edge Devices in Multi-Modal Vision Models*, Provisional Patent Application, Registration No. 52494, Docket No. 62941.2834, InComm Payments, 2024. Filed.

Role: Lead Inventor.

Innovation: A low-cost multimodal computer vision system for real-time customer behavior analysis through multi-view emotion, gaze, hand-interactions and product detection - deployed in edge device.

Research Prototypes

- **DockGraph:** Fast and interactive research prototype for graph-based protein-protein docking for near-native protein-protein pairs. [Link](#)
- **SpaceCondenseNet:** A web-based space protein research tool for predicting and visualizing how proteins may aggregate, phase-separate, and study the protein-protein interactions under microgravity-inspired transport conditions. [Link](#)

Peer Review

Paper reviewer at IEEE Internet of Things Journal. [Web of Science Profile](#)

Paper reviewer at 34th International Conference on Artificial Neural Networks (ICANN).

Paper reviewer at 37th International Conference on Tools with Artificial Intelligence (ICTAI).

Editorial Board

Session Chair: The 62nd ACMSE 2024 Conference. [Link](#)

Technical Program Committee Member: 37th IEEE International Conference on Tools with Artificial Intelligence (ICTAI 2025).

Academic Projects

LLM RAG Chatbot Development: Designed a retrieval-augmented generation chatbot integrating large language models with external knowledge bases.

- Leveraged LangChain and Neo4j for context-aware responses and structured data querying.
- Deployed backend with FastAPI and developed an interactive interface using Streamlit.
- Optimized RAG models for real-time data retrieval and conversational outputs.

Hierarchical Federated Learning with MultiTree All-Reduce: Developed a hierarchical federated learning framework integrating MultiTree all-reduce with Adaptive Hierarchical Federated Learning Process.

- Achieved $O(\log n)$ communication complexity compared to $O(n)$ ring all-reduce while maintaining comparable convergence.
- Reduced communication overhead to 3.70J per device and 10.86ms per all-reduce operation.
- Achieved 75%+ CIFAR-10 test accuracy with only 3-4% degradation under Non-IID data.

Awards and Achievements

- Received technical scholarship for undergraduate excellence. 2014 – 2015
- Earned Bangladesh-Sweden Trust Fund Scholarship. 2024
- Received travel grants from the university to attend conferences. 2024, 2025, 2026
- Earned a place in the College of Computing and Software Engineering 4.0 Club at Kennesaw State University for Fall 2023, Spring 2024, Summer 2024, and Spring 2025.
- Earned first prize in cybersecurity training at Guardian Life Insurance, Dhaka, Bangladesh. March, 2022

Leadership and Activities

- **President**, Bangladesh Student Association (BSA), KSU April 2025 – June 2026

- **Secretary**, IEEE Student Branch, KSU
- IEEE Member
- Volunteer, ACM Southeast Conference 2024.

May 2025 – June 2026

Nov 2024 – Present

December, 2023

Certifications

- Human Subjects Research (Social, Behavioral, & Educational), CITI Program, Kennesaw State University, Apr 2025 – Apr 2028. [Link](#)
- Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning, DeepLearning.AI. [Link](#)
- Fundamentals of Quantitative Modeling, University of Pennsylvania. [Link](#)